Anita Petrovic

[LinkedIn](#) | Geneva, Switzerland

Work Experience

CERN, Mechanical Engineer (Origin) — **September 2024 - Present**
Mechanical engineer working on FCC studies in the cryogenics department — *Geneva, Switzerland*

- Support FCC-ee cryogenic engineering for superconducting RF cryomodules, QRL/transfer lines, and surface/underground technical infrastructure.
- Sized helium bi-phase line and MCI burst-disk line concepts for RF cryomodules to control vapour speed, helium inventory/level stability, flow velocity, and pressure differentials.
- Develop Python models for QRL heat loads and cryogenic distribution properties. Including conduction/radiation, pressure drop, temperature/enthalpy evolution, mass-flow profiles, JT/flash behaviour, node mixing, and distributed heat inleaks.
- Define cryogenic requirements for utilities, electrical racks, building interconnections, installation strategy, and integration constraints
- Coordinate cryogenic design and integration activities with RF, EN/ACE, Integration, and Technical Infrastructure teams; serve as the CAD/PLM point of contact for FCC cryogenics.
- Prepared technical inputs, reports, and presentations for the FCC Feasibility Study Report, Cryogenics LEP3 Report, FCC Week, TE Graduates Workshop, and TE FCC-ee Workshop Day.

Lucid Motors, Chassis Mechanical Design Engineering Intern — **May 2022 - August 2022**
Luxury Electric Vehicle Company — *Newark, CA*

- Owned the design of a bushing fixture to test components of the Lucid Air Suspension system
- Used Catia FEA to limit fixture deflection and verified that the design could withstand a max load of 18.2 kN
- Created and executed a static and dynamic test plan with the CTW LA-48 actuator
- Wrote a MATLAB program to analyze test data from low-cost bushing suppliers to verify performance similarities for static and dynamic stiffness to current high-cost bushing suppliers
- Designed, prototyped, and tested string potentiometer fixtures to collect and analyze rear steering data
- Project involved setting up sensors and DAQ. VBOX and CANalyzer were used for data logging
- Developed a concept jig model for future rear steering testing of the Lucid Gravity SUV

Tesla, Structural Design Engineering Intern — **September 2021 - December 2021**
Electric Vehicle and Clean Energy Company — *Fremont, CA*

- Owned the redesign of SEMI structural roof components which achieved a 5.4% mass reduction while solving for stamping feasibility, mass manufacturing, durability, and component integration
- Independently designed and implemented new toe board features for Model S/X which reduced cost and weight by over 7% while retaining structural integrity for frontal and corner crashes
- Iterated designs for FDM 3D printed fixtures with feedback from factory technicians that addressed NVH issues and reduced application time of adhesives by 10.5%
- Updated GD&T for new product design changes for Model S/X

Projects

EcoCar, Mechanical Team Lead — **September 2018 - May 2023**
Shell Eco-marathon, Fuel Cell Vehicle Team - University of Alberta — *Edmonton, AB*

- Led mechanical work for a fuel-cell vehicle team, managing 12 members, build cycles, scrum sprint cycles, sponsorships, and a $25,000 annual budget.
- Modeled the fuel cell, supercapacitor, DC/DC converter, and motor system in Simulink; designed 2020 Urban-Concept suspension components with a 48% weight reduction.
- Managed a four-month fiberglass and carbon composite body layup operation and developed compact suspension/steering concepts for a confined monocoque.

Education

University of Alberta — **June 2023**
BSc Mechanical Engineering, GPA: 3.9/4.0 — *Edmonton, AB*

- Scovan Engineering Prize in Design Project Management

Skills

CAD/PLM: CATIA V5/3DX, 3DEXPERIENCE, SmarTeam, SOLIDWORKS Certified Professional (CSWP)
Analysis/Programming: Python, Matplotlib, MATLAB, Simulink, ANSYS, CATIA FEA, Excel
Engineering: Cryogenic systems, QRL/transfer-line modeling, heat-load estimation, pressure drop, GD&T, design for manufacturing, composite manufacturing, technical reporting
Training: Cryostat Engineering for Helium Superconducting Devices, Python Programming, Matplotlib, Clear Scientific Presentations, SmarTeam to PLM